# Isaac Mubiru

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## PROFESSIONAL SUMMARY

Full-stack engineer, **strong in JavaScript/TypeScript and Node.js** (5+ years in production), who builds **developer tools and engineering automation** for other engineers and integrates **AI coding agents into real workflows** with hard validation gates and human review. Built an agentic job-search pipeline on Claude Code that runs agents in scheduled workflows, verifies every result against ground truth before acting, and keeps a human on the final submit. Wrote the SDK documentation and Ops training a team onboarded from after leading a **zero-downtime migration of 20+ production apps**, the largest behaviour-preserving change the company had made. React and React Native on the client side. Code: github.com/zac-09

## CORE COMPETENCIES

JavaScript/TypeScript & Node.js (5+ yrs) | Developer Tooling & Engineering Automation | AI Coding Agents in Workflows | Output Validation & Human Review | Git, CI/CD & Release Workflows | Code Quality & Behaviour-Preserving Refactoring | React & React Native (Desktop/Mobile UIs) | Systematic Debugging

## WORK EXPERIENCE

### MTailor

Sep 2022 – Present

Software Engineer (US Remote, reports to CTO)

- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime** — behaviour-preserving change across every app, with a reversible step at each stage and rollback held open for the whole cutover window.
- Built **real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub**; designed consumers to be idempotent so replays and out-of-order delivery were safe under live traffic.
- Saved the company **$5,000/month** by migrating all services off AWS onto one platform.
- Built a 3D visualisation feature using video overlay and ffmpeg, increasing buyer conversion; implemented Express Shipping, adding **$40 revenue per order**.
- Migrated file storage from Amazon S3 to Google Cloud Storage via a Python script.

Stack: Node.js, JavaScript, Firebase/Firestore, MongoDB, GCP Pub/Sub, AWS, Python, ffmpeg, React, Git

### MTailor (Contractor)

Mar 2022 – Jul 2022

Backend Migration Contractor (US Remote)

- Wrote the **Node.js and Python SDK documentation fellow engineers onboarded from** and trained the Ops team on the new Firebase dashboard and backup procedures — tooling and enablement adopted by the whole team.
- Built the migration tooling reading from MongoDB and writing to Firestore after complex processing logic, used to move every application.
- Hired via Upwork for one migration; became the company-wide migration lead.

Stack: Node.js, JavaScript, Python, MongoDB, Firebase, GCP, AWS

### Dr Wealth

Aug 2021 – Jan 2022

Software Engineer (Singapore Remote)

- Extended backend APIs in **TypeScript** using Firebase Cloud Functions and Express.js hosted on Heroku.
- Built ad-hoc jobs querying over **2 million Firestore records** to keep customer-facing prices current from Morningstar APIs on a real-time PWA.
- Added responsive UI pages to the PWA using React, Tailwind and HTML.

Stack: Node.js, TypeScript, Firebase, Express.js, React, Tailwind

### Mind2matter

Jan 2022 – Jul 2022

Software Engineer (US Remote, reports to CTO)

- Built backends for DeFi applications using Web3 and Node.js for a US agency with demanding clients.
- Delivered React UIs from Figma designs under tight timelines.

Stack: Node.js, JavaScript, Web3, Firebase, React, Tailwind

### CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead (Uganda Remote)

- Led a team of 4 developers building legal-tech systems for NGOs (FIDA Uganda, LASPNET Uganda).
- Developed a cross-platform **mobile app with React Native and Expo (iOS & Android)** with Firebase push notifications, shipped to both stores.
- Architected a microservices backend using Apache Kafka, Docker and Kubernetes with gRPC for synchronous calls; independent containerised deploys per service.
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC.

Stack: Node.js, JavaScript, React Native, Expo, Apache Kafka, Docker, Kubernetes, gRPC, Firebase

## PROJECTS

### Career-ops — Agentic Engineering Pipeline on Claude Code

AI agents in scheduled workflows

Node.js automation that runs AI agents on a schedule to scan sources, verify each result against ground truth with Playwright before any agent is allowed to score it, deduplicate against history, log every rejection with a reason, and stop at a human-reviewed submit. Operating rules measured and enforced in config: a source that yields nothing twice is disabled with a dated note. Agent-written heuristics reviewed and tightened like any other PR.

Node.js, Claude Code, Playwright, YAML, Puppeteer, Git

### Healthcare WhatsApp Chatbot with OCR

LLM + vision in production

Integrated the OpenAI API and Google Vision for prescription OCR over the WhatsApp Business API, with model output validated before anything is shown to a patient.

Node.js, OpenAI API, Google Vision, WhatsApp Business API

### Parse → Firebase Platform Migration (MTailor)

20+ apps, zero downtime

End-to-end migration owned from plan through cutover and rollback. Per-app sequencing, reversible at every step, migration tooling plus SDK docs and Ops training so the team could operate the new platform. $5,000/month saving on completion.

Node.js, Python, MongoDB, Firestore, GCP, AWS

### Event-Driven Microservices Backend (FIDA Uganda)

Kafka + Kubernetes

Service boundaries around Kafka events, containerised with Docker and deployed on Kubernetes, gRPC for synchronous calls. Served web, mobile and USSD clients from one set of services.

Node.js, Apache Kafka, gRPC, Docker, Kubernetes

## EDUCATION

**B.Sc. Computer Engineering —** Makerere University — Aug 2018 – Sep 2022

Kampala, Uganda

## SKILLS

**Proficient:** Node.js, JavaScript, TypeScript, React, Firebase, GCP, Python, MongoDB, SQL, Git, Linux, HTML, CSS
**AI tooling:** Claude Code, agentic pipelines with Playwright verification, OpenAI API, Google Vision
**Delivery & infra:** Docker, Kubernetes, Apache Kafka, gRPC, Google Pub/Sub, Express.js, Expo, AWS
**Intermediate:** Go, Rust, React Native, Web3
**Languages:** English (fluent, 5+ yrs US/Singapore remote), Kiswahili